PAPER_232: NGC 1792 "The Stellar Forge" — Starburst Galaxy MUGE with Normalized SFR Mass Growth and SN-Driven Wind Feedback

Author: Daniel T. Murphy

NGC 1792, nicknamed "The Stellar Forge," is a starburst barred-spiral galaxy in the constellation Columba at $z = 0.0095$ *(~50 Mpc) with a specific star formation rate (sSFR) among the highest within 100 Mpc. This paper introduces two novel MUGE methods: (1) a normalized specific star formation rate amplitude* $SFR_{factor} = SFR_{M_\odot/yr} / M_{total,M_\odot}$ *applied to a time-evolving mass* $M(t) = M_0(1 + SFR_{factor} \cdot e^{-t/\tau_{SF}})$*, and (2) supernova-driven outflow feedback* $a_{SN} = \rho_{wind} v^2_{SN} / \rho_{fluid}$*. This system was* ***not previously represented*** *in the CP1/CP2/CP3 pipeline prior to Session 58.*

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Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok*share*8d951e12.txt extraction — Doc 19, PREVIOUSLY UNKNOWN SYSTEM) **Date:** March 2026 **Classification:** Novel MUGE — Specific SFR Normalization + SN-Driven Outflow Feedback **Status:** Proof-Quality Whitepaper

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Abstract

NGC 1792, nicknamed "The Stellar Forge," is a starburst barred-spiral galaxy in the constellation Columba at $z = 0.0095$ (~50 Mpc) with a specific star formation rate (sSFR) among the highest within 100 Mpc. This paper introduces two novel MUGE methods: (1) a normalized specific star formation rate amplitude $SFR_{factor} = SFR_{M_\odot/yr} / M_{total,M_\odot}$ applied to a time-evolving mass $M(t) = M_0(1 + SFR_{factor} \cdot e^{-t/\tau_{SF}})$, and (2) supernova-driven outflow feedback $a_{SN} = \rho_{wind} v^2_{SN} / \rho_{fluid}$. This system was **not previously represented** in the CP1/CP2/CP3 pipeline prior to Session 58.

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1. Physical System

NGC 1792 is a grand-design barred spiral with multiple starburst regions visible in infrared and $H\alpha$:

Parameter	Value
Constellation	Columba
Morphology	SAB(rs)bc barred spiral
Distance	~ 50 Mpc
z	0.0095
M_0	$10^{10} M_\odot$
r	$80{,}000$ ly
B	$5\ \mu$T
SFR	$10 M_\odot$/yr
τ_{SF}	100 Myr

Parameter	Value
SN wind: ρ_{wind}	10^{-21} kg/m^3
SN wind: v_{SN}	2000 km/s

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2. Novel Equations

2.1 Normalized Specific SFR Mass Growth

$$M(t) = M_0\left(1 + SFR_{factor} \cdot e^{-t/\tau_{SF}}\right)$$

where:

$$SFR_{factor} = \frac{SFR [M_\odot/\text{yr}]}{M_{total} [M_\odot]} = \frac{10}{10^{10}} = 10^{-9} \text{ yr}^{-1}$$

This normalization converts the star formation rate into a dimensionless fractional mass growth rate — the **specific SFR** — and uses it directly as the amplitude of the exponential mass evolution. This is mathematically distinct from all prior MUGE mass-growth implementations.

2.2 Canonical Value at t = 50 Myr

$$M(50\ \text{Myr}) = 10^{10}\left(1 + 10^{-9} \times e^{-50/100}\right) = 10^{10}(1 + 6.065 \times 10^{-10}) \approx 10^{10} M_\odot$$

The fractional change is minute ($\sim 10^{-9}$) — this is appropriate for a 10 Gyr old galaxy with $10^{10} M_\odot$ growing at 10 $M_\odot$/yr.

2.3 Supernova-Driven Outflow Feedback

$$a_{SN} = \frac{\rho_{wind} v^2_{SN}}{\rho_{fluid}}$$

With $\rho_{wind} = \rho_{fluid} = 10^{-21}$ kg/m^3 (SN ejecta same density as ambient medium at early stage):

$$a_{SN} = v^2_{SN} = (2 \times 10^6)^2 = 4 \times 10^{12} \text{ m/s}^2$$

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3. Starburst Context

NGC 1792's starburst is driven by interactions with neighboring galaxies (NGC 1792 group). Key observational properties:

- $H\alpha$ emission tracing $\sim 10 M_\odot$/yr SFR
- Infrared luminosity $L_{IR} \approx 3 \times 10^{10} L_\odot$
- SN rate: $\sim 0.1–0.3$ SN/century (consistent with $10 M_\odot$/yr SFR from Kroupa IMF)

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4. Previously Unknown Status

Prior to Session 58, NGC 1792 was absent from the CP1/CP2/CP3 library. As a starburst spiral in the low-redshift universe with well-measured SFR, it fills a gap between:

- Quiescent spirals (NGC 3596, NGC 2525)
- Merging galaxies (Antennae, PAPER_235)

High-z starburst (HUDF, PAPER_231)

5. Simulation Parameters

Parameter	Value
`t_{canonical}`	`50` Myr
`dt`	`0.5` Myr
`SFR_{factor}`	`10^{-9}` yr`^{-1}`
`\tau_{SF}`	`100` Myr
`H(z)`	Friedmann at `z=0.0095`

6. Calculator Class

```
class GalaxyNGC1792StarburstForgeCalculator(_CP3Calculator):
"""PAPER_232: NGC 1792 'Stellar Forge' — sSFR mass growth, SN wind feedback (PREVIOUSLY UNKNOWN)"""
# Session 58 — grok_share_8d951e12.txt Doc 19
```

Located in `CondensedPhysics3.py` (Session 58).

7. Conclusion

NGC 1792 introduces two novel MUGE methods: specific SFR normalization as a mass growth amplitude (`SFR_{factor} = SFR/M_{total}`) and SN-driven outflow feedback using the standard ram-pressure formula. As a previously unknown system in the CP pipeline, it fills the low-redshift starburst niche in the MUGE library.

Source: grok*share*8d951e12.txt — Doc 19 (NGC 1792 Stellar Forge Starburst MUGE, previously unknown system)

UQFF computed: MUGE buoyancy ratio U*bi*/*F*U = [SSq]■?■r■/GM = 5.7e-1■5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s■ at r_ISCO.